

R Programming – Student Performance Analysis

Dataset: Attendance, Assignment Score and Final Examination Marks

Student	Attendance (%)	Assignment Score	Marks
S1	70	65	60
S2	75	70	65
S3	80	75	72
S4	85	80	78
S5	90	90	88

1. Create the dataset

```
student <- c("S1", "S2", "S3", "S4", "S5")
attendance <- c(70, 75, 80, 85, 90)
assignment <- c(65, 70, 75, 80, 90)
marks <- c(60, 65, 72, 78, 88)

data <- data.frame(student, attendance, assignment, marks)
print(data)
```

Output: The data frame contains the five students and their attendance, assignment scores and final marks.

2. Calculate the mean of each variable

```
mean(attendance)
mean(assignment)
mean(marks)
```

Calculated results: Mean attendance = 80%, mean assignment score = 76, and mean final marks = 72.6.

3. Find the correlation between attendance and final marks

```
cor(attendance, marks)
```

The correlation is approximately 0.99, showing a very strong positive relationship: as attendance increases, final marks also tend to increase.

4. Find the correlation between assignment score and final marks

```
cor(assignment, marks)
```

The correlation is approximately 0.99, showing a very strong positive relationship between assignment performance and final examination marks.

5. Linear regression: Predict marks from attendance

```
model1 <- lm(marks ~ attendance, data = data)
summary(model1)

# Prediction for a student with 95% attendance
predict(model1, data.frame(attendance = 95))
```

The regression model estimates final marks from attendance. The positive slope indicates that higher attendance is associated with higher marks.

6. Linear regression: Predict marks from assignment score

```
model2 <- lm(marks ~ assignment, data = data)
```

```
summary(model2)

# Prediction for an assignment score of 85
predict(model2, data.frame(assignment = 85))
```

This model estimates final examination marks from assignment score. A positive relationship indicates that better assignment performance is associated with better final marks.

7. Scatter plot

```
plot(attendance, marks,
     main = "Attendance vs Final Marks",
     xlab = "Attendance (%)",
     ylab = "Final Examination Marks",
     pch = 19)

abline(model1)
```

The scatter plot visually shows the relationship between attendance and final examination marks. The regression line shows the overall increasing trend.

8. Complete R program

```
student <- c("S1", "S2", "S3", "S4", "S5")
attendance <- c(70, 75, 80, 85, 90)
assignment <- c(65, 70, 75, 80, 90)
marks <- c(60, 65, 72, 78, 88)

data <- data.frame(student, attendance, assignment, marks)
print(data)

cat("Mean Attendance:", mean(attendance), "\n")
cat("Mean Assignment Score:", mean(assignment), "\n")
cat("Mean Final Marks:", mean(marks), "\n")

cat("Correlation: Attendance vs Marks =",
    cor(attendance, marks), "\n")

cat("Correlation: Assignment vs Marks =",
    cor(assignment, marks), "\n")

model1 <- lm(marks ~ attendance, data = data)
print(summary(model1))

model2 <- lm(marks ~ assignment, data = data)
print(summary(model2))

plot(attendance, marks,
     main = "Attendance vs Final Marks",
     xlab = "Attendance (%)",
     ylab = "Final Examination Marks",
     pch = 19)

abline(model1)
```

Conclusion

The analysis indicates a strong positive relationship between attendance, assignment performance and final examination marks. Students with higher attendance and better assignment scores generally achieved higher final marks.